

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LABORATORY PERFORMANCE

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

4. Develop an AWT application containing two text fields and buttons for: ADD
CLEAR- When the ADD button is clicked, calculate and display the sum of two numbers.
When CLEAR is clicked, clear the input fields."

Test Cases:

tc1_input- Number 1=10, Number 2=20, Click ADD

tc1_output- Sum = 30

tc2_input- Number 1=100, Number 2=250, Click ADD

tc2_output- Sum = 350

tc3_input- Number 1=-10, Number 2=20, Click ADD

tc3_output- Sum = 10

CODE:

A. JAVA CODE

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class AdditionAWT extends Frame implements ActionListener {
```

```
    Label number1Label, number2Label, resultLabel;
```

```
    TextField number1Field, number2Field, resultField;
```

```
    Button addButton, clearButton;
```

```
public AdditionAWT() {

    setTitle("Addition Calculator");

    setSize(450, 300);

    setLayout(new GridLayout(4, 2, 10, 10));

    number1Label = new Label("Number 1:");
    number1Field = new TextField();

    number2Label = new Label("Number 2:");
    number2Field = new TextField();

    resultLabel = new Label("Result:");
    resultField = new TextField();
    resultField.setEditable(false);

    addButton = new Button("ADD");
    clearButton = new Button("CLEAR");

    addButton.addActionListener(this);
    clearButton.addActionListener(this);

    add(number1Label);
    add(number1Field);
```

```
add(number2Label);
```

```
add(number2Field);
```

```
add(resultLabel);
```

```
add(resultField);
```

```
add(addButton);
```

```
add(clearButton);
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent e) {  
        System.exit(0);  
    }  
});
```

```
setVisible(true);
```

```
}
```

```
public void actionPerformed(ActionEvent e) {
```

```
    if (e.getSource() == addButton) {
```

```
        try {
```

```
            double number1 = Double.parseDouble(number1Field.getText());
```

```
            double number2 = Double.parseDouble(number2Field.getText());
```

```

        double sum = number1 + number2;

        if (sum == (int) sum) {
            resultField.setText("Sum = " + (int) sum);
        } else {
            resultField.setText("Sum = " + sum);
        }

    } catch (NumberFormatException ex) {
        resultField.setText("Invalid Input");
    }
}

if (e.getSource() == clearButton) {
    number1Field.setText("");
    number2Field.setText("");
    resultField.setText("");
}
}

public static void main(String[] args) {
    new AdditionAWT();
}
}

```

OUTPUT:

 Addition Calculator

Number 1:	50
Number 2:	289
Result	Sum = 339
ADD	CLEAR